

MCP Model Context Protocol

1. What is MCP?

MCP stands for Model Context Protocol. It is an open standard created by Anthropic (released late 2024) that gives AI models a clean, consistent way to connect to outside tools and data sources.

Think of it like USB-C for AI. Just like USB-C lets you plug any device into any laptop, MCP lets any AI model talk to any external service without custom code every time.

Before MCP, every AI app needed its own one-off connector for every tool it used. MCP fixes this by giving everyone one standard way to do it.

Why it matters

- AI models alone only know what they were trained on and have no live data access
- MCP bridges them to databases, APIs, and files in real time
- Uses JSON-RPC 2.0 as the communication standard (lightweight and widely supported)
- By 2026: 10,000+ public MCP servers and 97M+ monthly SDK downloads

2. How MCP Works

Every MCP setup has three pieces:

Component	What it is	Example
MCP Host	The AI app you are using	Claude, Gemini CLI, VS Code
MCP Client	Lives inside the host, handles the connection	The part of Claude that talks to servers
MCP Server	Exposes a service's tools and data to the AI	BigQuery server, Maps server

3. What an MCP Server Exposes

An MCP server exposes exactly three things:

- **Tools** - Let the AI do things. Call an API, run a query, send a message. Something actually happens.
- **Resources** - Let the AI read data without changing anything. Like reading a file or schema.
- **Prompts** - Pre-built input templates that help users interact with a tool effectively.

This simple 3-thing structure keeps MCP lean but covers almost every real use case.

Local vs Remote MCP Servers

- Local servers run on your machine, use standard input/output to communicate
- Remote servers run on the provider's cloud and expose an HTTP endpoint
- Remote servers are more reliable, easier to manage, and enterprise-ready

4. Google's MCP Support

Google announced fully-managed remote MCP servers in December 2025. Instead of you figuring out how to connect an AI to BigQuery or Maps, Google just gives you an endpoint. Point your agent at it and you are done.

Google Services Available via MCP

- **Google Maps** - Location data, routing, weather, nearby places
- **BigQuery** - Query enterprise data directly, data stays secure in place
- **Google Compute Engine** - AI can provision and resize infrastructure automatically
- **Google Kubernetes Engine** - AI manages containers, fixes issues, optimizes costs
- Coming soon: Cloud Run, Cloud SQL, Spanner, Looker, Cloud Logging, SecOps, and more

5. Google's Security Features for MCP

This is where Google stands out. They do not just give you access, they give you access with full enterprise controls.

i. IAM (Identity and Access Management)

- Only verified agents, clients, and users can authenticate and use MCP tools
- Fine-grained policies: you decide who can do what on which resource
- IAM deny policies let admins block specific MCP tool usage entirely
- Fully compliant with the official MCP authorization specification

ii. Model Armor

- Google's AI security layer that sits between the agent and the MCP server
- Scans every MCP call going in and every response coming out
- Protects against prompt injection, sensitive data leaks, and tool poisoning
- Floor settings: baseline safety filters that nobody can turn off
- Currently in Preview as of 2026

iii. Audit Logging

- Every MCP tool call is logged automatically for full traceability
- Integrated with Google Cloud Trace for detailed monitoring
- You can see exactly what your AI agent did and when

iv. VPC Service Controls

- Locks MCP access to inside your private network perimeter
- Prevents data from leaving your secure zone

v. Discovery and Governance

- Cloud API Registry: central place to find trusted MCP tools from Google
- Apigee API Hub: lets your org publish and govern its own APIs as MCP tools
- Security Command Center: threat detection specifically for AI agents and MCP servers

6. Google MCP Security at a Glance

Security Feature	What it does
IAM	Controls who can access which MCP tools and how
Model Armor	Scans calls and responses for injection and data leaks
Audit Logging	Records every single MCP tool call automatically
VPC Service Controls	Keeps MCP usage inside a private network
Cloud API Registry	Central place to discover trusted MCP servers
Security Command Center	Threat detection for AI agents and MCP servers

7. Key Terms to Remember

- **MCP** - Open standard for connecting AI to external tools and data
- **MCP Server** - Program that exposes a service's tools or data to an AI
- **MCP Host** - The AI app that uses those tools (e.g. Claude, Gemini)
- **MCP Client** - The bridge inside the host that talks to servers
- **Model Armor** - Google's shield that checks AI inputs and outputs for threats
- **Tool Poisoning** - Attack where a bad MCP server tricks an AI into doing harmful things
- **Prompt Injection** - Attack where hidden instructions in data hijack the AI's behavior